

NEW PLYMOUTH, New Zealand

WMO: 933090

Lat: 39.0078S

Lon: 174.1836E

Elev: 92

StdP: 14.65

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	36.2	38.2	31.7	26.2	38.9	33.7	28.4	40.9	34.4	53.7	32.0	53.3	7.0	140	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.1	74.8	66.5	73.0	65.9	71.5	65.1	69.1	72.2	67.8	70.9	66.6	69.9	11.7	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.0	103.2	70.7	66.6	98.2	69.6	65.3	93.8	68.7	33.3	72.2	32.2	71.0	31.3	70.1	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 29.9	26.7	24.0	DB	31.7	79.1	1.8	2.8	30.4	81.1	29.4	82.7	28.4	84.3	27.1	86.3	(4)
(5)			WB	30.6	70.8	1.7	1.8	29.3	72.1	28.3	73.1	27.4	74.1	26.2	75.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.7	63.5	64.6	62.2	58.5	55.0	51.3	49.7	50.4	52.7	54.5	57.0	61.0	(6)	
	DBStd	6.30	3.87	3.54	3.82	4.01	4.36	4.50	3.99	3.60	3.54	3.43	3.53	3.74	(7)	
	HDD50	168	0	0	0	1	9	39	57	40	16	6	0	0	(8)	
	HDD65	3135	74	46	103	196	309	412	475	452	370	324	242	132	(9)	
	CDD50	2593	419	407	377	257	165	78	47	53	95	146	209	340	(10)	
	CDD65	88	29	34	15	2	0	0	0	0	0	0	0	8	(11)	
	CDH74	94	36	41	9	0	0	0	0	0	0	0	1	7	(12)	
	CDH80	1	1	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	11.6	11.3	10.6	10.9	10.5	11.3	11.8	11.6	11.4	12.4	13.3	12.8	11.6	(14)	
(15) Precipitation	PrecAvg	56.7	4.0	3.7	4.0	4.6	5.4	5.9	6.0	5.2	4.8	4.8	4.5	4.2	(15)	
	PrecMax	79.4	9.2	15.7	14.3	13.3	10.3	10.6	11.7	10.9	9.8	17.6	9.5	7.5	(16)	
	PrecMin	44.1	0.6	0.2	0.5	0.9	0.4	2.0	0.4	2.1	1.2	0.7	1.7	0.8	(17)	
	PrecStd	7.2	2.2	2.7	2.6	2.7	1.9	1.9	2.6	2.0	2.2	2.9	2.0	1.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.4	77.9	75.2	70.5	66.5	62.0	60.0	60.1	62.9	65.2	70.5	73.8	(19)	
		MCWB	68.2	68.0	65.7	63.7	61.7	59.2	56.8	55.8	57.1	59.5	62.0	65.5	(20)	
	2%	DB	74.5	75.1	72.7	68.2	64.7	60.5	58.5	58.6	60.6	62.8	67.2	71.0	(21)	
		MCWB	66.7	66.8	64.9	62.8	61.1	57.8	55.7	55.0	56.6	58.5	61.0	64.0	(22)	
	5%	DB	72.6	73.3	71.0	66.9	63.4	59.5	57.5	57.5	59.4	61.5	65.1	69.3	(23)	
		MCWB	66.0	66.4	64.4	62.0	60.3	56.8	54.5	54.3	56.0	57.6	60.1	63.2	(24)	
	10%	DB	70.8	71.7	69.4	65.6	62.2	58.4	56.4	56.6	58.5	60.3	63.4	67.6	(25)	
		MCWB	65.0	65.6	63.6	61.1	59.2	55.6	53.3	53.5	55.4	56.8	59.0	62.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.7	71.1	68.9	66.5	63.8	60.7	58.5	58.0	59.3	61.0	64.4	68.2	(27)	
		MCDB	74.4	74.0	71.6	68.3	64.8	61.4	59.1	58.8	60.4	63.0	67.5	70.6	(28)	
	2%	WB	68.7	69.5	67.4	64.9	62.5	58.9	56.7	56.5	58.3	59.7	62.6	66.3	(29)	
		MCDB	72.1	72.3	70.2	66.7	63.7	60.0	58.0	57.6	59.5	61.6	65.5	68.9	(30)	
	5%	WB	67.3	68.3	66.1	63.7	61.2	57.5	55.3	55.4	57.3	58.7	61.3	65.0	(31)	
		MCDB	70.7	71.6	69.4	65.8	63.0	59.1	56.9	56.9	58.6	60.6	63.8	67.8	(32)	
	10%	WB	66.1	66.9	64.8	62.1	59.6	56.1	53.8	54.1	56.3	57.8	60.1	63.8	(33)	
		MCDB	69.7	70.5	68.3	64.8	61.8	58.1	55.9	56.1	58.0	59.8	62.5	66.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.4	13.1	13.5	12.8	11.4	10.9	11.1	11.1	10.7	10.5	11.4	11.2	(35)	
		MCDBR	14.2	14.2	14.5	13.2	10.5	8.7	10.3	10.9	11.0	11.2	13.1	13.2	(36)	
		MCWBR	9.1	8.9	9.3	9.3	8.3	7.6	8.6	8.5	8.7	8.8	9.2	8.7	(37)	
	5% WB	MCDBR	11.3	11.0	10.9	10.0	7.8	7.3	8.3	8.5	8.5	8.9	10.1	10.0	(38)	
		MCWBR	8.1	8.2	8.3	8.4	7.3	7.3	8.0	7.7	7.7	7.9	8.3	7.6	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.347	0.342	0.325	0.313	0.305	0.300	0.295	0.309	0.335	0.338	0.335	0.351	(40)	
	taud		2.540	2.575	2.621	2.646	2.636	2.626	2.638	2.599	2.507	2.512	2.538	2.529	(41)	
	Ebn at Noon		312	306	299	283	266	256	266	280	290	304	314	312	(42)	
	Edh at Noon		34	32	28	24	21	19	20	24	30	33	34	35	(43)	
(44) All-Sky Solar Radiation	RadAvg		2182	1910	1531	1033	691	539	624	875	1221	1613	1986	2067	(44)	
	RadStd		129	170	76	74	36	39	47	53	88	113	154	141	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days					
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)			
(46)	Station Only	N/A	N/A	+0.77	N/A	N/A	N/A	N/A	-160	N/A	N/A	(46)		
(47)	Regional (4 neighbors)	+0.52	+0.90	+1.10	N/A	N/A	N/A	-56	-187	+142	N/A	(47)		

Nomenclature: See separate page